SOFTWARE / TOOLS REQUIRED:

- MATLAB

Theory:-

Butterworth Lowpass Filters

This filter is characterized by the property that its magnitude is flat in both passband and stopband. The magnitude squared response of an Nth-order lowpass filter is given by

$$|H_a(j\Omega)|^2 = \frac{1}{1 + (\frac{\Omega}{\Omega_c})^{2N}}$$

where N is the order of the filter and Ωc is the cutoff frequency in rad/sec

Chebyshev Lowpass

Filters The magnitude-squared response of a Chebyshev-I filter is

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 T_N^2(\frac{\Omega}{\Omega_c})}$$

where N is the order of the filter, is the passband ripple factor, which is related to Rp, and TN(x) is the Nth-order Chebyshev polynomial given by :-

$$\mathrm{T}_N(x) = \begin{cases} \cos(N\cos^{-1}(x)), & 0 \le x \le 1 \\ \cosh(\cosh^{-1}(x)), & 1 < x < \infty, \text{where } x = \frac{\Omega}{\Omega_c} \end{cases}$$

The Equiripple response of the Chebyshev filters is due to this polynomial T_N(x). Its key properties are

(a) for 0 < x < 1, T_N(x) Oscillates between-1 and 1, and

(b) for 1 < x < ∞,T_N(X)increases monotonically to ∞

Design Procedure for impulse invariable Technique

Given the digital lowpass filter specifications ωp, ωs, Rp and As, we want to determine H(s) by first designing an equivalent analog filter and then mapping it into the desired digital filter. The steps required for this procedure are 1. Choose T and determine the analog frequencies

$$\Omega_p = \frac{\omega_p}{T_p} \quad \text{and} \quad \Omega_s = \frac{\omega_s}{T}$$

2. Design an analog filter Ha(s) using the specifications Ωp, Ωs, Rp and As. This can be done using anyone of the three (Butterworth, Chebyshev, or elliptic) prototypes of the previous section.

3. Using partial fraction expansion, expand Ha(s) into

$$H_a(s) = \sum_{k=1}^{N} \frac{R_k}{s - p_k}$$

Design Procedure for Bi-linear transformation Technique

Given the digital lowpass filter specifications ωp, ωs, Rp and As, we want to determine H(z). The design steps in the procedure are the following:

1. Choose T. This is arbitrary, and we may set T = 1.

2. Prewrap the cutoff frequencies ωp and ωs; that is, calculate Ωp and Ωs,

$$\Omega_p = \frac{2}{T} tan\left(\frac{\omega_p}{2}\right), \ \Omega_s = \frac{2}{T} tan\left(\frac{\omega_s}{2}\right)$$

3. Design an analog filter Ha(s) using the specifications Ωp, Ωs, Rp and As.

4. Finally, set

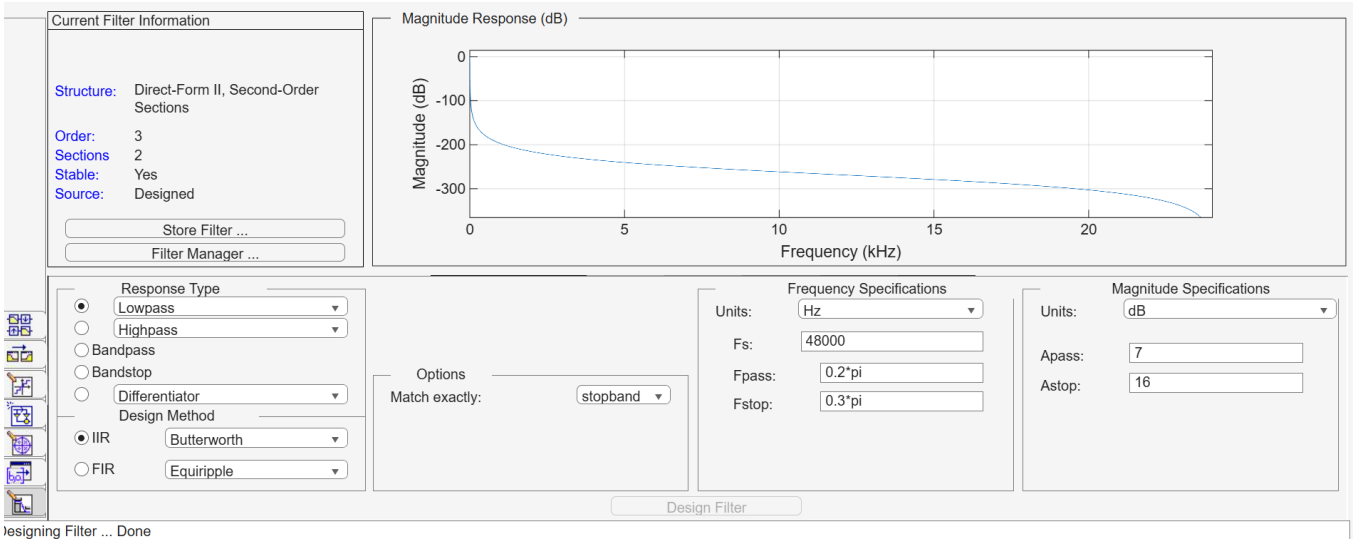
$$H(z) = H_s\left(\frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}}\right)$$

<u>Aim 1</u>:- Design a lowpass Butterworth filter to satisfy passband cutoff: Ωp = 0.2π; Passband ripple: Rp = 7 dB Stopband cutoff: Ωs = 0.3π; Stopband ripple: As = 16 dB

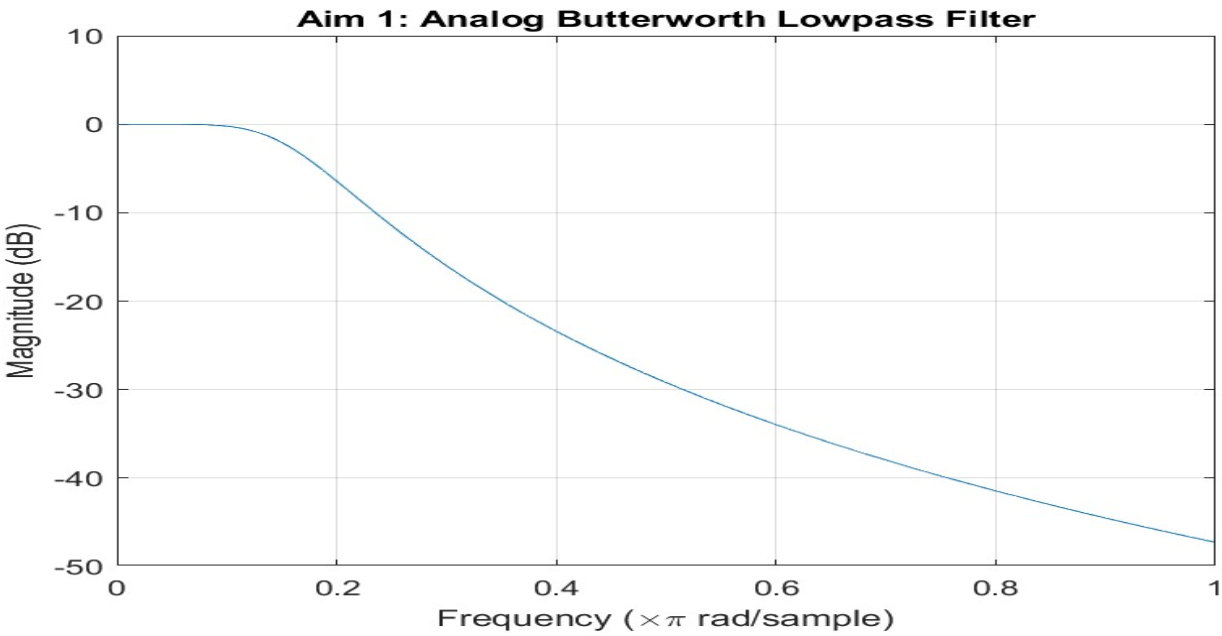
Code:-

```
% Specifications

Wp = 0.2*pi; Rp = 7;

Ws = 0.3*pi; As = 16;


% 1. Find Order and Cutoff Frequency

[N, Wc] = buttord(Wp, Ws, Rp, As, 's');


% 2. Generate Transfer Function

[num, den] = butter(N, Wc, 's');

H = tf(num, den);


% 3. Plot Magnitude Response

w = 0:0.01:pi;

h = freqs(num, den, w);

plot(w/pi, 20*log10(abs(h)));

grid on;

xlabel('Frequency (\times\pi rad/sample)');

ylabel('Magnitude (dB)');

title('Aim 1: Analog Butterworth Lowpass Filter');
```

Output :-



FILTER DESIGNER



Conclusion :-

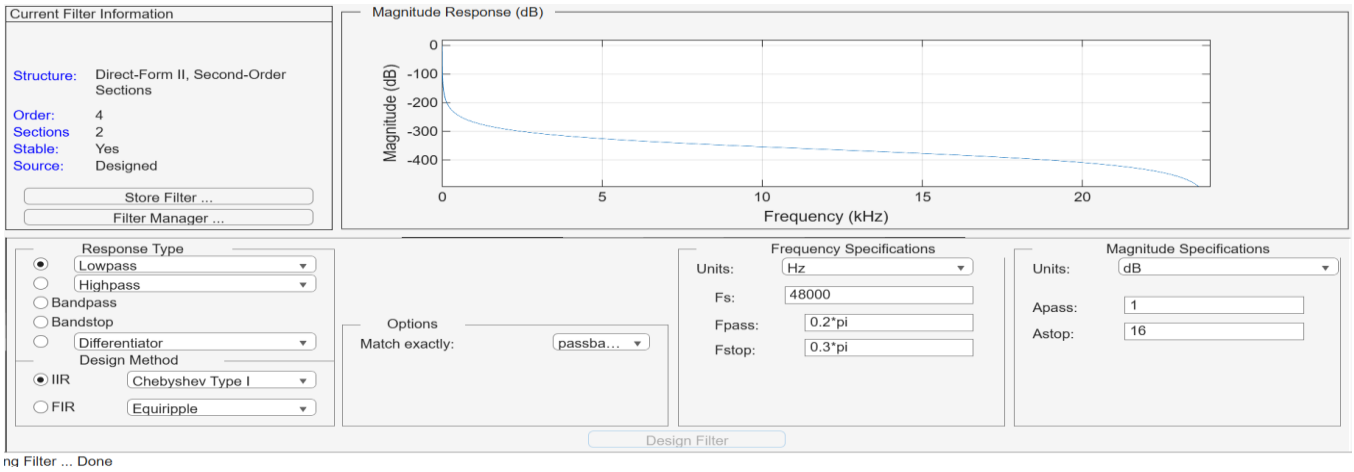
The Butterworth lowpass filter was designed to meet the given passband and stopband specifications. The required filter order was obtained using buttord, ensuring a smooth, maximally flat response in the passband. The plotted magnitude response verifies that the filter satisfies the desired attenuation and cutoff requirements

<u>Aim 2</u>:- Design a lowpass Chebyshev-I filter to satisfy passband cutoff: Ωp = 0.2π; Passband ripple: Rp = 1 dB Stopband cutoff: Ωs = 0.3π;Stopband ripple: As = 16 dB

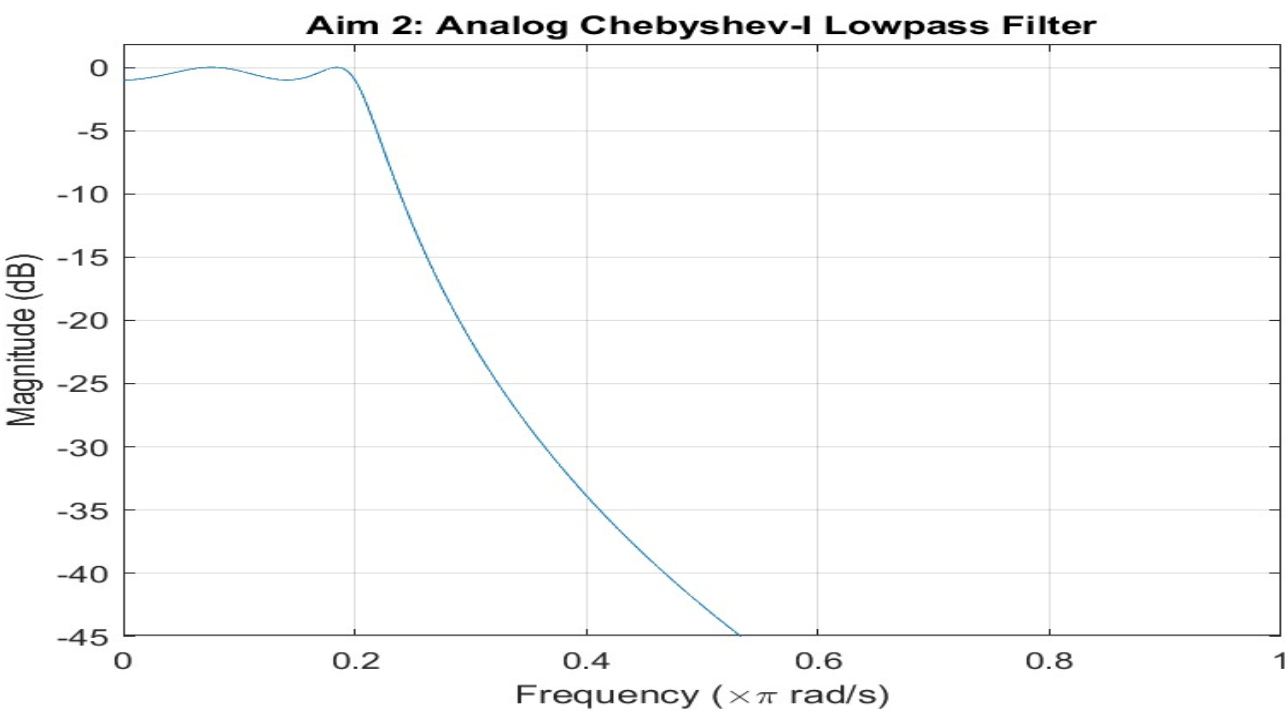
<u>CODE:-</u>

```
Wp = 0.2*pi;

Rp = 1;

Ws = 0.3*pi;

As = 16;


[N, Wp_actual] = cheb1ord(Wp, Ws, Rp, As, 's');

[num, den] = cheby1(N, Rp, Wp_actual, 's');


w = linspace(0, pi, 1000);

h = freqs(num, den, w);


figure;

plot(w/pi, 20*log10(abs(h)));

grid on;

xlabel('Frequency (\times\pi rad/s)');

ylabel('Magnitude (dB)');

title('Aim 2: Analog Chebyshev-I Lowpass Filter');

axis([0 1 -30 2]);
```

Output :-



FILTER DESIGNER



Conclusion :-

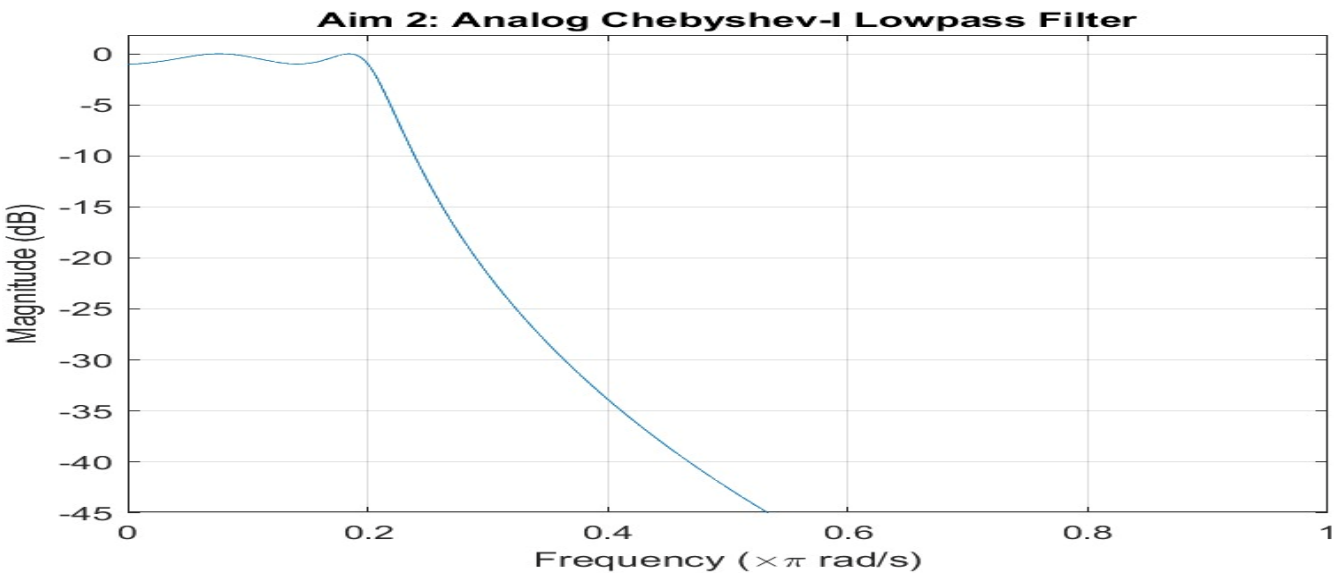
The Chebyshev-I lowpass filter was designed to meet the specified passband and stopband requirements with a smaller passband ripple of 1 dB. Using cheb1ord, the minimum filter order was obtained, resulting in a sharper transition compared to the Butterworth filter. The frequency response shows controlled ripples in the passband and faster attenuation in the stopband, confirming that the designed filter satisfies the given specifications effectively.

Aim 3: Design a lowpass digital filter (Butterworth) using impulse invariance technique to satisfy ωp = 0.2π, Rp = 1 dB , ωs = 0.3π,As = 15 dB

CODE:-

```
wp = 0.2*pi;

ws = 0.3*pi;

Rp = 1;

As = 15;

T = 1;


Wp = wp/T;

Ws = ws/T;


[N, Wc] = buttord(Wp, Ws, Rp, As, 's');


[num_a, den_a] = butter(N, Wc, 's');


[num_d, den_d] = impinvar(num_a, den_a, 1/T);


[H, w] = freqz(num_d, den_d, 1024);


figure;

plot(w/pi, 20*log10(abs(H)));

grid on;

xlabel('Normalized Frequency (\times\pi rad/sample)');

ylabel('Magnitude (dB)');

title('Aim 3: Digital Butterworth Filter (Impulse Invariance)');

axis([0 1 -40 5]);
```

Output :-



Conclusion :-

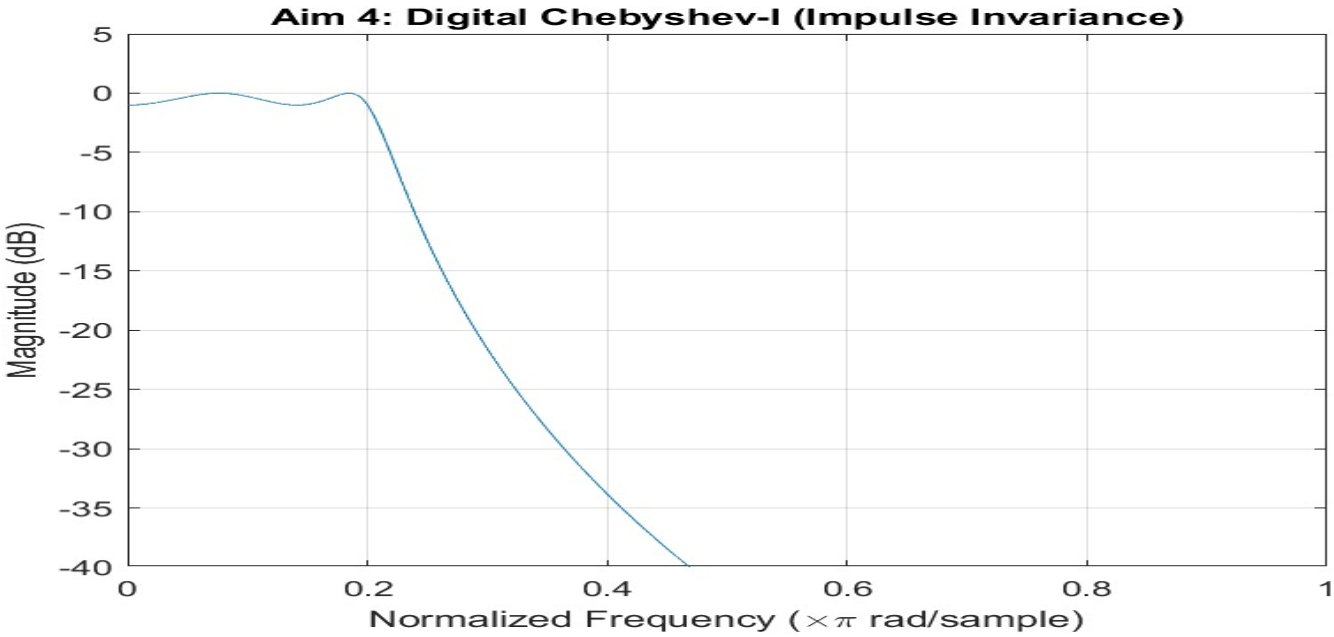
The digital Butterworth lowpass filter was successfully designed using the impulse invariance technique to meet the given specifications. The analog prototype was first obtained and then converted into a digital filter, preserving its characteristics. The frequency response shows a smooth, ripple-free passband and adequate attenuation in the stopband, confirming that the designed filter satisfies the required performance criteria.

<u>Aim 4:</u> Design a lowpass digital filter (Chebyshev-I) using impulse invariance technique to satisfy ωp = 0.2π,Rp = 1 dB ωs = 0.3π,As = 15 dB

<u>CODE</u>:-

```
wp = 0.2*pi;

ws = 0.3*pi;

Rp = 1;

As = 15;

T = 1;


Wp = wp/T;

Ws = ws/T;

[N, Wp_actual] = cheb1ord(Wp, Ws, Rp, As, 's');

[num_a, den_a] = cheby1(N, Rp, Wp_actual, 's');

[num_d, den_d] = impinvar(num_a, den_a, 1/T);

[H, w] = freqz(num_d, den_d, 1024);


figure;

plot(w/pi, 20*log10(abs(H)));

grid on;

xlabel('Normalized Frequency (\times\pi rad/sample)');

ylabel('Magnitude (dB)');

title('Aim 4: Digital Chebyshev-I (Impulse Invariance)');

axis([0 1 -40 5]);
```

Output :-



Conclusion :-

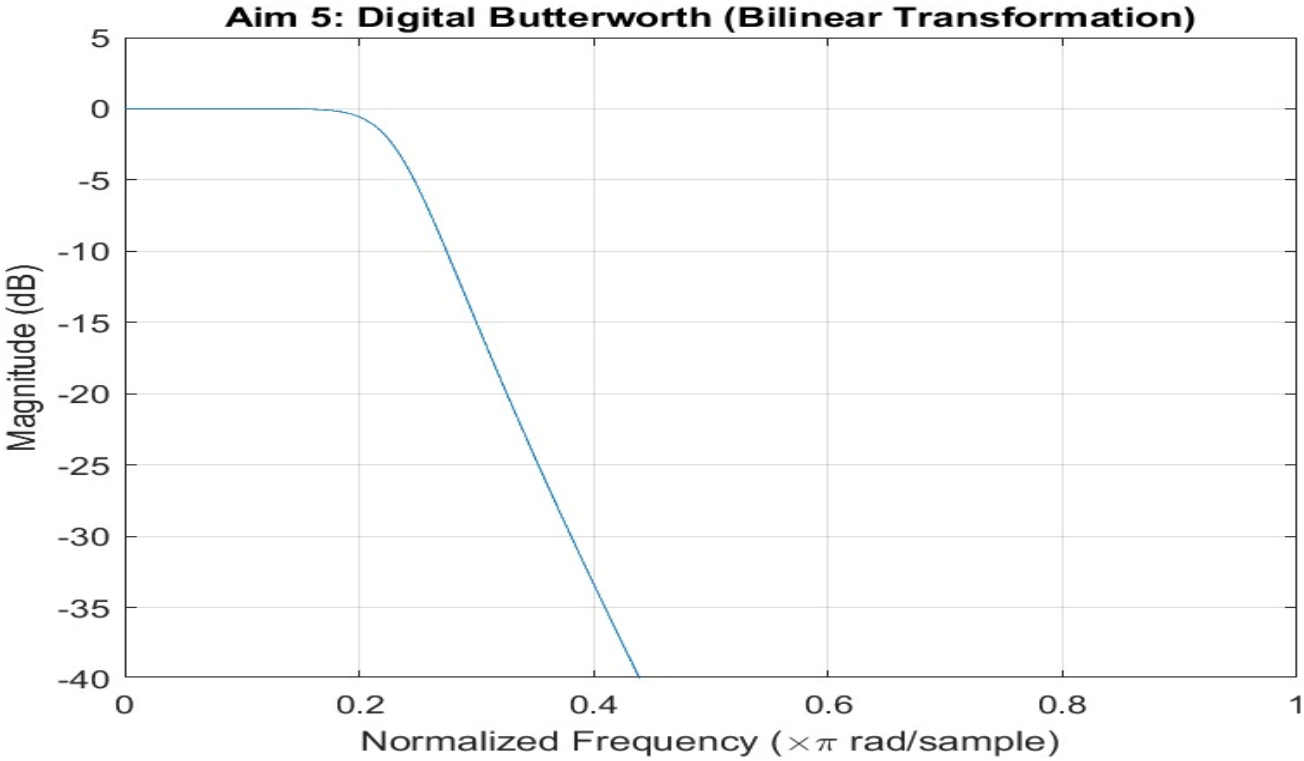
The digital Chebyshev-I lowpass filter was designed using the impulse invariance technique to satisfy the given specifications. The analog prototype with controlled passband ripple was converted into a digital filter, resulting in a sharper transition compared to the Butterworth design. The frequency response confirms the presence of passband ripples and sufficient stopband attenuation, demonstrating that the filter meets the required design criteria effectively.

<u>Aim 5:</u> Design a digital filter (Butterworth) using Bilinear transformation technique to satisfy ωp = 0.2π, Rp = 1 dB, ωs = 0.3π, As = 15 dB

<u>CODE</u>:-

```
wp = 0.2*pi;

ws = 0.3*pi;

Rp = 1;

As = 15;

T = 1;


Wp = (2/T) * tan(wp/2);

Ws = (2/T) * tan(ws/2);

[N, Wc] = buttord(Wp, Ws, Rp, As, 's');

[num_a, den_a] = butter(N, Wc, 's');

[num_d, den_d] = bilinear(num_a, den_a, 1/T);

[H, w] = freqz(num_d, den_d, 1024);


figure;

plot(w/pi, 20*log10(abs(H)));

grid on;

xlabel('Normalized Frequency (\times\pi rad/sample)');

ylabel('Magnitude (dB)');

title('Aim 5: Digital Butterworth (Bilinear Transformation)');

axis([0 1 -40 5]);
```

Output :-



Conclusion :-

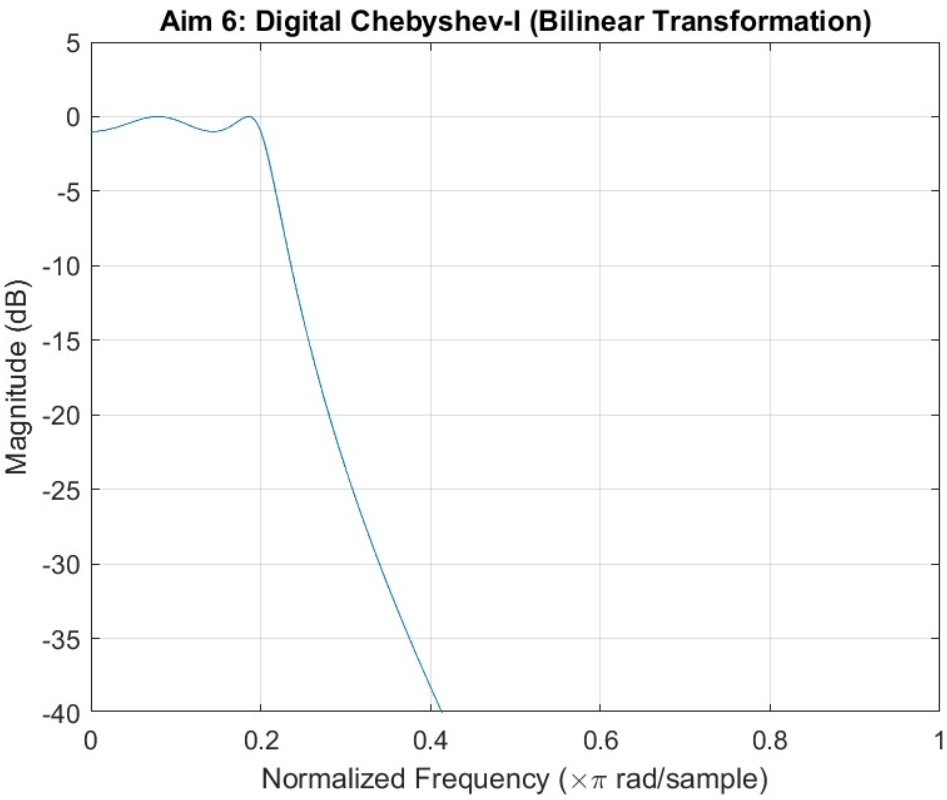
The digital Butterworth lowpass filter was designed using the bilinear transformation technique to meet the specified requirements. Frequency pre-warping was applied to accurately map digital frequencies to the analog domain, avoiding aliasing effects. The resulting filter exhibits a smooth, ripple-free passband and adequate stopband attenuation, and the frequency response confirms that the design satisfies the given specifications effectively

<u>Aim 6:</u> Design a digital filter (Chebyshev-I) using Bilinear transformation technique to satisfy ωp = 0.2π,Rp = 1 dB ωs = 0.3π,As = 15 dB

<u>CODE:-</u>

```
wp = 0.2*pi;

ws = 0.3*pi;

Rp = 1;

As = 15;

T = 1;


Wp = (2/T) * tan(wp/2);

Ws = (2/T) * tan(ws/2);


[N, Wp_actual] = cheb1ord(Wp, Ws, Rp, As, 's');


[num_a, den_a] = cheby1(N, Rp, Wp_actual, 's');


[num_d, den_d] = bilinear(num_a, den_a, 1/T);


[H, w] = freqz(num_d, den_d, 1024);


figure;

plot(w/pi, 20*log10(abs(H)));

grid on;

xlabel('Normalized Frequency (\times\pi rad/sample)');

ylabel('Magnitude (dB)');

title('Aim 6: Digital Chebyshev-I (Bilinear Transformation)');

axis([0 1 -40 5]);
```

Output :-



Conclusion :-

The digital Chebyshev-I lowpass filter was designed using the bilinear transformation technique to meet the given specifications. Pre-warping ensured accurate frequency mapping from the digital to analog domain, eliminating aliasing effects. The resulting filter exhibits controlled ripples in the passband and a sharper transition to the stopband, and the frequency response confirms that the design satisfies the required attenuation and cutoff criteria effectively.